VIDHI VATS

Software Engineer | Backend & Full-Stack | Node.js · Python · Distributed Systems

+91 9906843553 · vidhii.vats@gmail.com · linkedin.com/in/vidhivats- · github.com/vidhi-vats-11 · vidhivats-portfolio.vercel.app

Meerut, Uttar Pradesh · Available to join immediately

SUMMARY

Software engineer with 9 months of production experience building real-time distributed data systems in Python and Django — ingestion pipelines, Redis caching, RabbitMQ queues and WebSocket delivery on GCP — plus AI and network-security tooling. Independently designed and shipped a full e-commerce platform on Node.js, Express, Prisma and PostgreSQL with JWT auth, inventory reservation and order workflows. Comfortable owning a service end to end: schema, API, caching, deployment. Open to full-time and internship roles; available to join immediately.

EXPERIENCE

Software Developer Intern — **Data Sailors Private Limited** 06/2025 – 02/2026

Python, Django, MongoDB, Redis, RabbitMQ, WebSockets, React.js, LangChain, GCP, Docker, Raspberry Pi

- Built backend services in Django + MongoDB that ingest and process high-frequency telemetry from a 12-node distributed sensor network.
- Introduced a RabbitMQ queue to decouple ingestion from processing and Redis caching for hot reads, absorbing traffic spikes that previously degraded dashboard responsiveness under load.
- Delivered live updates over WebSockets, replacing polling and cutting the delay between a sensor event and its appearance on the React.js operator dashboard, which I built to give a single view across all 12 nodes.
- Built a Retrieval-Augmented Generation assistant over an internal document corpus — chunking, embedding generation and vector indexing, with source citations to keep answers grounded; containerised with Docker and deployed through GitHub Actions.
- Developed a network vulnerability scanner in Python + Django performing real-time device discovery, and tested SSH/Telnet credential strength against simulated MITM and ARP-spoofing conditions.
- Engineered a Data Concentrator Unit on Raspberry Pi for secure collection and transmission of data from edge devices to the cloud.
- Deployed and operated the stack on Google Cloud Platform with Docker and secure remote access over VPN + SSH; scaled the architecture from a single node to a distributed cluster, improving throughput and fault tolerance.

PROJECTS

Thrift Vault — production e-commerce platform (live · code)

Node.js, Express, Prisma, PostgreSQL, Redis, JWT, Zod, React, Vercel

- Designed a 14-table relational schema in Prisma/PostgreSQL covering catalog, variants, cart, wishlist, orders, payments and reviews.
- Built a modular Express API across 9 domains with JWT access/refresh authentication, bcrypt password hashing, Zod request validation, Helmet hardening and Redis-backed rate limiting.
- Implemented stock reservation at checkout with a scheduled job that releases inventory from expired orders to prevent overselling of one-of-one items, and abstracted payments behind a gateway interface so a live provider can drop in without touching order logic.

Personal Portfolio — responsive personal site (live · code)

React, Vite, JavaScript, EmailJS, GitHub Actions, Vercel

- Built and deployed a responsive portfolio with an EmailJS-powered contact form and an interactive resume chatbot.
- Automated build and deployment through a GitHub → Vercel CI/CD pipeline with environment-based secret management.

TECHNICAL SKILLS

Backend	Node.js, Express.js, Django, Flask, REST APIs, JWT Auth, Prisma ORM
Languages	Python, JavaScript, TypeScript, C++, SQL
Data & Messaging	PostgreSQL, MongoDB, MySQL, Redis, RabbitMQ
Cloud & DevOps	GCP, AWS, Docker, GitHub Actions, CI/CD, Vercel, Linux
Frontend	React.js, Tailwind CSS, HTML, CSS — **AI / LLM:** LangChain, OpenAI APIs, RAG, Embeddings, Vector Stores
Foundations	Data Structures & Algorithms, OOP, System Design, Distributed Systems

EDUCATION

M.Tech, Computer Science & Engineering — **Indian Institute of Technology (IIT) Patna** 2026 – 2028

B.Tech, Electrical Engineering — **Shri Mata Vaishno Devi University** 2021 – 2025

CERTIFICATION

Full Stack Generative & Agentic AI with Python — Udemy: LLMs, AI agents, RAG and automation pipelines.